

Introduction to Nanotechnology

1. Introduction

Nanotechnology refers to science and engineering at the nanoscale. Materials with dimensions between 1 and 100 nanometers are called nanomaterials. Nanomaterials behave differently from bulk materials because of their extremely small size.

2. Surface-to-Volume Ratio

As particle size decreases, the surface-to-volume ratio increases significantly. A larger percentage of atoms remain on the surface of the material. This changes the chemical reactivity and physical properties of nanomaterials.

3. Quantum Confinement

Quantum confinement affects the electronic behavior of nanomaterials. The density of states changes when dimensions become extremely small. This results in unique optical and electrical properties.

4. Applications of Nanotechnology

Nanotechnology is used in medicine, electronics, energy storage, and sensors. Nanoparticles are used in drug delivery and cancer therapy. Nanomaterials are also used in solar cells and water purification systems.

5. Comparison Table

Property	Bulk Material	Nanomaterial
Surface Area	Low	High
Reactivity	Normal	Enhanced
Optical Properties	Stable	Size-dependent